26	Find the ripple frequency of a full-wave rectifier that operates off a 50 Hz power line	25	50	100	200	1	C
27	Find Vout for the below op-amp with input offset voltage of 5mV and open loop gain of 10000	15V	-5mV	-50V	12V	1	D
28	2's complement representation of a 16-bit number is FFFF. Its magnitude in decimal is	0	1	32767	65535	1	B
29	In the below circuit if the speed of the motor is to be increased where do you place the resistor R2(100)?	A	B	C	D	1	B
30	If R = 4 ohms and if the current through the resistor X is 0A, what is the value of current through the resistor R?	1A	0.5 A	0.75 A	Data inadequate	1	B
31	A steady voltage of 1V applied to the circuit at t=0. The current through the resistor R just after t = 0 and at steady state are	100mA, 50mA	50mA, 50mA	100mA, 100mA	50mA, 100mA	1	A
32	What are the values of Rc , Re and Rb in the figure shown below?	2.2 K, 1.2 K, 356K	22 K, 12 K, 356 K	220 K, 120 K, 356 K	120 K, 220 K, 356 K	1	A
3	In the below circuit, the silicon transistor has a very large amplification factor. If Ic = 1mA, calculate R2.	10 Kohm	20 Kohm	30 Kohm	40 Kohm	1	A
4	What is the logic function of below circuit	NAND	XNOR	NOR	XOR	1	B
	Determine the output Y	Y = AB	Y = /(AB)	Y = A + B	Y = /(AB) + /(AB)	1	A

Version A

Question ID	Question -------- SET-A	Option 1	Option 2	Option 3	Option 4	Mark	Question Answer
1	A 10V reference is drawn from the circuit shown in the figure. Zener diode of 10V, 400mW and with firing current of 5mA is used. The values of Rs is	50 ohms	100 ohms	200 ohms	0 ohms	1	B
2	What is the equivalent resistance for the circuit shown below	Equal to 6	Less than 6	Greater than 6	None of the above	1	B
3	An ideal current meter & volt meter should have	Infinite resistance & Zero resistance	Finite resistance & Infinite resistance	Infinite resistance & Finite resistance	Zero resistance & Infinite resistance	1	D
4	What is the voltage on capacitor C2 when all three switches are turned on?	16V	20V	12V	10V	1	D
5	What will be the output of Not gate if we give Sine wave (3v pp with 1.5v dc base) ?:	Sine wave ranging from –Vcc/2 to +Vcc/2	Square wave ranging from –Vcc/2 to +Vcc/2	Sine wave ranging from 0V to +Vcc	Square wave ranging from 0V to +Vcc	1	D
6	When transistors are used in digital circuits they usually operate in the:	active region	breakdown region	saturation and cutoff regions	linear region	1	C
7	What is the output voltage across the 900 ohm in the circuit given below	10V	14.67V	20V	9.47V	1	A
8	If the voltage applied across a capacitance is triangular in waveform then the waveform of the current is	Trapezoidal	Triangular	Rectangular	Sinusoidal	1	C
9	The number of comparators present in a 4 bit flash ADC is	4	5	15	16	1	C
10	What is the capacitive reactance of below circuit?	0	Infinite	1	0.0125	1	B

Version A

Question ID	Question -------- SET-A	Option 1	Option 2	Option 3	Option 4	Mark	Question Answer
1	A 10V reference is drawn from the circuit shown in the figure. Zener diode of 10V, 400mW and with firing current of 5mA is used. The values of Rs is	50 ohms	100 ohms	200 ohms	0 ohms	1	B
2	What is the equivalent resistance for the circuit shown below	Equal to 6	Less than 6	Greater than 6	None of the above	1	B
3	An ideal current meter & volt meter should have	Infinite resistance & Zero resistance	Finite resistance & Infinite resistance	Infinite resistance & Finite resistance	Zero resistance & Infinite resistance	1	D
4	What is the voltage on capacitor C2 when all three switches are turned on?	16V	20V	12V	10V	1	D
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9	The number of comparators present in a 4 bit flash ADC is	4	5	15	16	1	C
10	What is the capacitive reactance of below circuit?	0	Infinite	1	0.0125	1	B

Version A

11	For the inputs shown in the below figure, the output Y is	0	1	AB	Complement of AB	1	A
12	What is the fall time of the wave shown in the figure?	30	40	16	20	1	C
13	Find the voltage across 9ohm resistor	9 V	6 V	3 V	Circuit incomplete	1	C
14	With a Triangular wave at the input, the circuit given below produces	Sine wave	Cosine wave	Rectangular wave	None of the above	1	C
15	With a square wave given at the input, the circuit shown below produces	Sine wave	Triangular wave	Square wave	None	1	B
16	What is the output of the below circuit	0	1	Clock	HIZ	1	C
17	The output of the circuit shown below with sine wave input is	Fig A	Fig B	Fig C	Fig D	1	A
18	D1, D2 & D3 are LEDs in the circuit. Which of them will glow now?	D1, D2	D2, D3	D1, D3	D3 only	1	C
19	What is the output pk-pk voltage in below circuit with Si diodes?	7.4 V	6.7 V	11. 7 V	6.4 V	1	A
20	Find the minimum and maximum Zener diode current.	11mA, 24mA	5mA, 5mA	1mA, 9mA	0mA, 24mA	1	C
21	What is the Logic Function realized by this circuit?	NOR	AND	NAND	OR	1	B
22	If the inputs 1 & 2 are given to a digital logic EX-OR gate, what will be the appropriate output wave shape?	W1	W2	W3	W4	1	D
23	Identify the logic operation of the below circuit.	NAND	NOR	XOR	AND	1	C
24	Identify the frequency response curve for a band-pass filter	Fig 1	Fig 2	Fig 3	Fig 4	1	A
25	What is the Output when Input is 0V or 3.3V?	3.3V, 0V	3.3V, 3.3V	0V,3.3V	Non deterministic, 0V	1	D

Version A

26	Find the ripple frequency of a full-wave rectifier that operates off a 50 Hz power line	25	50	100	200	1	C
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28	2's complement representation of a 16-bit number is FFFF. Its magnitude in decimal is	0	1	32767	65535	1	B
29	In the below circuit if the speed of the motor is to be increased where do you place the resistor R2(100)?	A	B	C	D	1	B
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32	What are the values of Rc , Re and Rb in the figure shown below?	2.2 K, 1.2 K, 356K	22 K, 12 K, 356 K	220 K, 120 K, 356 K	120 K, 220 K, 356 K	1	A
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35	Determine the output Y	Y = AB	Y = /(AB)	Y = A + B	Y = /(AB) + /(AB)	1	A

Version A

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31	A steady voltage of 1V applied to the circuit at t=0. The current through the resistor R just after t = 0 and at steady state are	100mA, 50mA	50mA, 50mA	100mA, 100mA	50mA, 100mA	1	A
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34	What is the logic function of below circuit	NAND	XNOR	NOR	XOR	1	B
35	Determine the output Y	Y = AB	Y = /(AB)	Y = A + B	Y = /(AB) + /(AB)	1	A

19. A circle has a diameter of 0.02 cm. What is the area of the circle in μm^2

20. Provide logical function of Y in terms of P and Q for below circuit

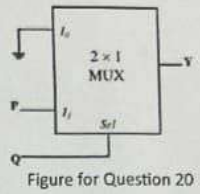
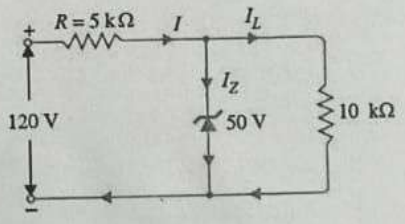


Figure for Question 20

PART B

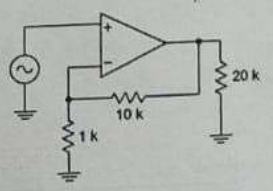
QUESTIONS (Answer any 4. Each question carries 5 marks)

1. What is Nyquist theorem? A system has a sampling rate of 50,000 samples per second. The maximum frequency of the signal it can acquire to reconstruct is?
2. a) Calculate the slip in an induction motor 4 pole 50 Hz rotating at 1400 RPM.
   b) A three phase, 50Hz, 6 pole induction motor runs at 960 rpm. The stator copper loss, core loss, and the rotational loss of the motor can be neglected. Calculate the percentage efficiency of the motor
3. A parallel plate capacitor has a capacitance of C. If the distance between the plates is reduced by half, and the space between the plates is filled with a medium having dielectric constant 6, Calculate the value of new capacitance in terms of 'C'.
4. A coil consists of 2000 tons of Copper (Cu) wire having a cross-sectional area of 0.8 mm^2. The mean length per turn is 80 cm. The resistivity of Cu is 0.02μΩ-m. Find the resistance of the coil and power absorbed by the coil when connected across 110 V DC supply.
5. For the circuit shown in figure, find
   i. Output Voltage
   ii. Voltage drop across series resistance
   iii. Current through Zener diode.

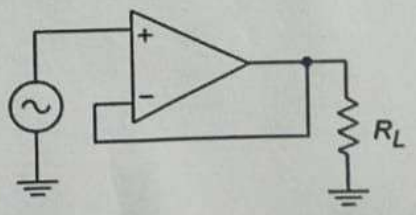


6. A current if 5A flows in a resistor of resistance 8 Ω. Determine
   i) The rate of heat dissipation
   ii) The total heat dissipated in 30s
7. Calculate the voltage gain of following circuits:

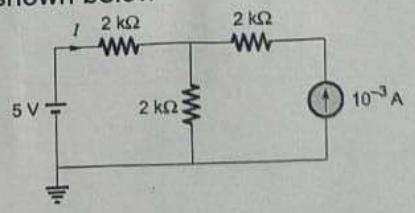
(a)



(b)



8. Calculate the I in the circuit shown below



QUESTION PAPER

ELECTRICAL, ELECTRONICS & ALLIED ENGINEERING DISCIPLINES

TOTAL MARKS = 50 | Time: 60 minutes

PART A

QUESTIONS (Answer any 15. Each question carries 2 marks)

1. What is the difference between Analog and Digital Signal?
2. What will be the power dissipation across a silicon diode carrying a current of 50mA?
3. How many conditions are required to properly realize a two-input system?
4. What is regenerative breaking?
5. Define power factor. Should it be high / low? Why?
6. Define Kirchoff's Current Law (KCL). How many junctions are present in the following circuit.

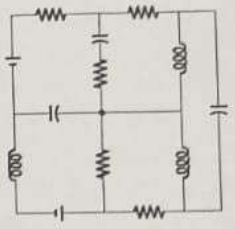
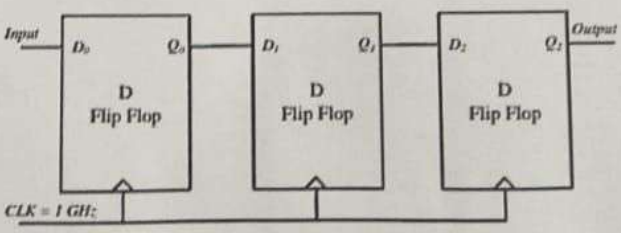


Figure for Question 6

7. What is IGBT? Among IGBT, SCR, Power MOSFET and GTO which has fastest switching characteristics?
8. What is meant by input offset voltage of an Op-Amp? What is CMRR?
9. Write the truth table and Boolean equation for XNOR gate. Also draw its symbol.
10. What is High-Impedance State in Tri-State Logic and Why it is important in digital circuits
11. What are the transformer losses?
12. What is the difference between sensors and actuators? Given one example of each.
13. What is the significance of voltage in selection of a capacitor?
14. What is the difference between isolator and circuit breaker?
15. What is the purpose of the transformer?
16. What is the difference between microprocessor and microcontroller?
17. What are buck converter and boost converters?
18. The synchronous sequential circuit shown below works a clock frequency of 1 GHz. What is circuit throughput, in Mbits/s? What is the circuit latency, in ns?